University of Florida, Department of Mathematics

Topology PhD Exam, May 2010

Work the following problems and show all work. Support all statements to the best of your ability.

1. Compute $\pi_1(S^2)$ and $H_*(S^2)$.

2. Let X be a connected completely regular topological space having more than one point. Can X be countable?

3. Does there exist a covering space of the figure eight with nontrivial abelian fundamental group?

4. Show that the 2-dimensional sphere with four points deleted cannot be a topological group.

5. Prove that there is no map of degree two from S^2 to the torus T^2.

Answer the following with complete definitions or statements or short proofs.

6. State the Baire Category Theorem.

7. State the homology Mayer–Vietoris Theorem.

8. State the Urysohn Lemma.

9. State the Five Lemma.

10. Compute the Euler characteristic $\chi(\mathbb{RP}^4 \times S^3 \times T^2)$.

11. Define retraction and deformation retraction.

12. State the Lefschetz Fixed Point Theorem.

13. Can the set of irrational numbers be presented as a countable union of closed in $\mathbb{R}$ subsets?

14. Draw a picture of the universal cover of the 2-sphere with the segment joining the north and south poles.

15. Describe all connected subsets of $\mathbb{R}$.